

P02 - Analytics Engineering Platform

Billing + customer data | source contracts -> staging -> dimensional marts -> tests -> incremental/history -> orchestration

Business/domain

Subscriptly recurring billing

How this book works

This book has exactly 13 phase pages. Each phase has one artifact/evidence contract and one protected rubric page. The small teaching pattern is not the final solution; apply the reasoning to the supplied project workspace yourself.

Project control before Phase 1

- Keep protected Review and sealed failure/incident packs closed until the named phase.
- Create a real project branch/working copy if using Git.
- Record tool/runtime versions only after observing them.
- Do not delete a bad first attempt.
- Use the exact phase evidence ID and path; this keeps Mentor/book/evidence routing one-to-one.

Phase map

P02-01 - Read the analytics-engineering brief and define outputs

P02-02 - Profile billing/customer sources and state grains

P02-03 - Design facts, dimensions, keys and serving grain

P02-04 - Create dbt project, sources and dependency graph

P02-05 - Build thin staging models with explicit contracts

P02-06 - Build dimensional marts without fan-out

P02-07 - Add data tests and critical quality gates

P02-08 - Implement incremental model/update behavior

P02-09 - Preserve history with snapshot/SCD reasoning

P02-10 - Add orchestration and reproducible run order

P02-11 - Inject duplicate/relationship/partial-run failures

P02-12 - Document lineage, runbook, limitations and evidence

P02-13 - Clean-clone build and analytics-platform defense

P02-01 - Read the analytics-engineering brief and define outputs

STARTING STATE

Open the project workspace and your last saved evidence. Keep protected Review and any sealed incident/failure pack closed unless this phase explicitly unlocks it.

OUTCOME

Independently complete read the analytics-engineering brief and define outputs and produce reviewable evidence that proves the result, not merely the action.

TEACHER EXPLANATION

Define serving outputs and consumers before models. Analytics engineering starts from trustworthy contracts, not a folder of SQL files.

TINY PATTERN - NOT YOUR FINAL SOLUTION

Toy: finance needs invoice-grain revenue plus day-region mart. Name grain and reconciliation for both.

BUILD THIS ARTIFACT

business_questions.md + assumptions.md

VALIDATE

Define expected invariant before execution, run task, compare actual vs expected, and save one independent control check.

CHANGED / FAILURE TEST

PHASE CHECK P02-01: one changed-input or broken-assumption test must be run and documented before mastery.

EVIDENCE + REVIEW ROUTE

Evidence ID: M18-P02-01-EVIDENCE. Save:

03_PROJECT_WORKSPACES/P02_ANALYTICS_ENGINEERING_PLATFORM/evidence/P02-01/attempt.json. P02 rubric -> exact phase criteria; review only after attempt evidence exists.

MASTER / MOVE-ON

L3 independent professional; no unresolved critical failure; reproducible artifact/evidence.

STOP - EXPLAIN BACK

Without reading this page, explain what changed in the system, what evidence proves it, one way it could be wrong, and what your changed/failure test taught you. Next: P02-02.

P02-02 - Profile billing/customer sources and state grains

STARTING STATE

Open the project workspace and your last saved evidence. Keep protected Review and any sealed incident/failure pack closed unless this phase explicitly unlocks it.

OUTCOME

Independently complete profile billing/customer sources and state grains and produce reviewable evidence that proves the result, not merely the action.

TEACHER EXPLANATION

Billing and customer sources have different grains. History changes customer attributes over time; invoice facts remain events.

TINY PATTERN - NOT YOUR FINAL SOLUTION

Profile invoice_id uniqueness, customer key coverage, status values and customer change records separately.

BUILD THIS ARTIFACT

source_profile/ + grain_key_defects.md

VALIDATE

Record row counts, grain, keys, null/duplicate/schema defects; independently sample-check at least one finding.

CHANGED / FAILURE TEST

Replay the same batch/event plus one late/changed record; detect duplicate/logical-state corruption if implementation is unsafe.

EVIDENCE + REVIEW ROUTE

Evidence ID: M18-P02-02-EVIDENCE. Save:

03_PROJECT_WORKSPACES/P02_ANALYTICS_ENGINEERING_PLATFORM/evidence/P02-02/attempt.json. P02 rubric -> exact phase criteria; review only after attempt evidence exists.

MASTER / MOVE-ON

L3 independent professional; no unresolved critical failure; reproducible artifact/evidence.

STOP - EXPLAIN BACK

Without reading this page, explain what changed in the system, what evidence proves it, one way it could be wrong, and what your changed/failure test taught you. Next: P02-03.

P02-03 - Design facts, dimensions, keys and serving grain

STARTING STATE

Open the project workspace and your last saved evidence. Keep protected Review and any sealed incident/failure pack closed unless this phase explicitly unlocks it.

OUTCOME

Independently complete design facts, dimensions, keys and serving grain and produce reviewable evidence that proves the result, not merely the action.

TEACHER EXPLANATION

Design facts/dimensions from process and grain. A dimension key on the one side must be unique for the version/join rule you choose.

TINY PATTERN - NOT YOUR FINAL SOLUTION

Write: one row in fct_revenue = one accepted invoice. Then decide how customer history joins without fan-out.

BUILD THIS ARTIFACT

p02_phase_artifact + evidence note

VALIDATE

Define expected invariant before execution, run task, compare actual vs expected, and save one independent control check.

CHANGED / FAILURE TEST

PHASE CHECK P02-03: one changed-input or broken-assumption test must be run and documented before mastery.

EVIDENCE + REVIEW ROUTE

Evidence ID: M18-P02-03-EVIDENCE. Save:

03_PROJECT_WORKSPACES/P02_ANALYTICS_ENGINEERING_PLATFORM/evidence/P02-03/attempt.json. P02 rubric -> exact phase criteria; review only after attempt evidence exists.

MASTER / MOVE-ON

L3 independent professional; no unresolved critical failure; reproducible artifact/evidence.

STOP - EXPLAIN BACK

Without reading this page, explain what changed in the system, what evidence proves it, one way it could be wrong, and what your changed/failure test taught you. Next: P02-04.

P02-04 - Create dbt project, sources and dependency graph

STARTING STATE

Open the project workspace and your last saved evidence. Keep protected Review and any sealed incident/failure pack closed unless this phase explicitly unlocks it.

OUTCOME

Independently complete create dbt project, sources and dependency graph and produce reviewable evidence that proves the result, not merely the action.

TEACHER EXPLANATION

The dependency graph should encode meaning: sources -> thin staging -> reusable dimensions/facts -> marts. Avoid circular or hidden dependencies.

TINY PATTERN - NOT YOUR FINAL SOLUTION

Draw ref/source edges before SQL. Every model gets owner/grain and test responsibility.

BUILD THIS ARTIFACT

models/SQL/dbt + lineage + row-count controls

VALIDATE

State output grain; test key/cardinality; reconcile critical measures to an independent control query.

CHANGED / FAILURE TEST

PHASE CHECK P02-04: one changed-input or broken-assumption test must be run and documented before mastery.

EVIDENCE + REVIEW ROUTE

Evidence ID: M18-P02-04-EVIDENCE. Save:

03_PROJECT_WORKSPACES/P02_ANALYTICS_ENGINEERING_PLATFORM/evidence/P02-04/attempt.json. P02 rubric -> exact phase criteria; review only after attempt evidence exists.

MASTER / MOVE-ON

L3 independent professional; no unresolved critical failure; reproducible artifact/evidence.

STOP - EXPLAIN BACK

Without reading this page, explain what changed in the system, what evidence proves it, one way it could be wrong, and what your changed/failure test taught you. Next: P02-05.

P02-05 - Build thin staging models with explicit contracts

STARTING STATE

Open the project workspace and your last saved evidence. Keep protected Review and any sealed incident/failure pack closed unless this phase explicitly unlocks it.

OUTCOME

Independently complete build thin staging models with explicit contracts and produce reviewable evidence that proves the result, not merely the action.

TEACHER EXPLANATION

Staging should be thin and explicit: rename/type/standardize, preserve source meaning, avoid business aggregations that hide lineage.

TINY PATTERN - NOT YOUR FINAL SOLUTION

Turn raw billing strings into typed columns while keeping invoice grain; document required/optional fields.

BUILD THIS ARTIFACT

source_contracts.md

VALIDATE

State output grain; test key/cardinality; reconcile critical measures to an independent control query.

CHANGED / FAILURE TEST

Inject one duplicate dimension/reference key or one-to-many fan-out risk and prove the validation catches it.

EVIDENCE + REVIEW ROUTE

Evidence ID: M18-P02-05-EVIDENCE. Save:

03_PROJECT_WORKSPACES/P02_ANALYTICS_ENGINEERING_PLATFORM/evidence/P02-05/attempt.json. P02 rubric -> exact phase criteria; review only after attempt evidence exists.

MASTER / MOVE-ON

L3 independent professional; no unresolved critical failure; reproducible artifact/evidence.

STOP - EXPLAIN BACK

Without reading this page, explain what changed in the system, what evidence proves it, one way it could be wrong, and what your changed/failure test taught you. Next: P02-06.

P02-06 - Build dimensional marts without fan-out

STARTING STATE

Open the project workspace and your last saved evidence. Keep protected Review and any sealed incident/failure pack closed unless this phase explicitly unlocks it.

OUTCOME

Independently complete build dimensional marts without fan-out and produce reviewable evidence that proves the result, not merely the action.

TEACHER EXPLANATION

Marts must preserve intended grain across joins. Fan-out is a correctness failure even when totals look plausible.

TINY PATTERN - NOT YOUR FINAL SOLUTION

Inject duplicate C202 dimension row and prove the join cardinality changes before trusting revenue totals.

BUILD THIS ARTIFACT

models/SQL/dbt + lineage + row-count controls

VALIDATE

State output grain; test key/cardinality; reconcile critical measures to an independent control query.

CHANGED / FAILURE TEST

Inject one duplicate dimension/reference key or one-to-many fan-out risk and prove the validation catches it.

EVIDENCE + REVIEW ROUTE

Evidence ID: M18-P02-06-EVIDENCE. Save:

03_PROJECT_WORKSPACES/P02_ANALYTICS_ENGINEERING_PLATFORM/evidence/P02-06/attempt.json. P02 rubric -> exact phase criteria; review only after attempt evidence exists.

MASTER / MOVE-ON

L3 independent professional; no unresolved critical failure; reproducible artifact/evidence.

STOP - EXPLAIN BACK

Without reading this page, explain what changed in the system, what evidence proves it, one way it could be wrong, and what your changed/failure test taught you. Next: P02-07.

P02-07 - Add data tests and critical quality gates

STARTING STATE

Open the project workspace and your last saved evidence. Keep protected Review and any sealed incident/failure pack closed unless this phase explicitly unlocks it.

OUTCOME

Independently complete add data tests and critical quality gates and produce reviewable evidence that proves the result, not merely the action.

TEACHER EXPLANATION

Quality gates should block publication for critical failures and warn for noncritical conditions. Each rule needs an owner/action.

TINY PATTERN - NOT YOUR FINAL SOLUTION

Unique invoice_id and relationships are blocking; a late-but-valid record may be warning depending on SLA.

BUILD THIS ARTIFACT

tests/ + reconciliation report + quarantine evidence

VALIDATE

Define expected invariant before execution, run task, compare actual vs expected, and save one independent control check.

CHANGED / FAILURE TEST

PHASE CHECK P02-07: one changed-input or broken-assumption test must be run and documented before mastery.

EVIDENCE + REVIEW ROUTE

Evidence ID: M18-P02-07-EVIDENCE. Save:

03_PROJECT_WORKSPACES/P02_ANALYTICS_ENGINEERING_PLATFORM/evidence/P02-07/attempt.json. P02 rubric -> exact phase criteria; review only after attempt evidence exists.

MASTER / MOVE-ON

L3 independent professional; no unresolved critical failure; reproducible artifact/evidence.

STOP - EXPLAIN BACK

Without reading this page, explain what changed in the system, what evidence proves it, one way it could be wrong, and what your changed/failure test taught you. Next: P02-08.

P02-08 - Implement incremental model/update behavior

STARTING STATE

Open the project workspace and your last saved evidence. Keep protected Review and any sealed incident/failure pack closed unless this phase explicitly unlocks it.

OUTCOME

Independently complete implement incremental model/update behavior and produce reviewable evidence that proves the result, not merely the action.

TEACHER EXPLANATION

Incremental models need a stable key, bounded change filter and late-update policy. Exact rerun must be logically idempotent.

TINY PATTERN - NOT YOUR FINAL SOLUTION

Process a new invoice/change twice and show only intended rows/history versions change.

BUILD THIS ARTIFACT

state ledger/watermark + rerun/backfill evidence

VALIDATE

State output grain; test key/cardinality; reconcile critical measures to an independent control query.

CHANGED / FAILURE TEST

Replay the same batch/event plus one late/changed record; detect duplicate/logical-state corruption if implementation is unsafe.

EVIDENCE + REVIEW ROUTE

Evidence ID: M18-P02-08-EVIDENCE. Save:

03_PROJECT_WORKSPACES/P02_ANALYTICS_ENGINEERING_PLATFORM/evidence/P02-08/attempt.json. P02 rubric -> exact phase criteria; review only after attempt evidence exists.

MASTER / MOVE-ON

L3 independent professional; no unresolved critical failure; reproducible artifact/evidence.

STOP - EXPLAIN BACK

Without reading this page, explain what changed in the system, what evidence proves it, one way it could be wrong, and what your changed/failure test taught you. Next: P02-09.

P02-09 - Preserve history with snapshot/SCD reasoning

STARTING STATE

Open the project workspace and your last saved evidence. Keep protected Review and any sealed incident/failure pack closed unless this phase explicitly unlocks it.

OUTCOME

Independently complete preserve history with snapshot/scd reasoning and produce reviewable evidence that proves the result, not merely the action.

TEACHER EXPLANATION

SCD/snapshot history preserves the truth that attributes change. Never overwrite old business state if historical reporting depends on it.

TINY PATTERN - NOT YOUR FINAL SOLUTION

C202 SMB -> Enterprise should create a new version with non-overlapping validity, not rewrite the past.

BUILD THIS ARTIFACT

p02_phase_artifact + evidence note

VALIDATE

Define expected invariant before execution, run task, compare actual vs expected, and save one independent control check.

CHANGED / FAILURE TEST

PHASE CHECK P02-09: one changed-input or broken-assumption test must be run and documented before mastery.

EVIDENCE + REVIEW ROUTE

Evidence ID: M18-P02-09-EVIDENCE. Save:

03_PROJECT_WORKSPACES/P02_ANALYTICS_ENGINEERING_PLATFORM/evidence/P02-09/attempt.json. P02 rubric -> exact phase criteria; review only after attempt evidence exists.

MASTER / MOVE-ON

L3 independent professional; no unresolved critical failure; reproducible artifact/evidence.

STOP - EXPLAIN BACK

Without reading this page, explain what changed in the system, what evidence proves it, one way it could be wrong, and what your changed/failure test taught you. Next: P02-10.

P02-10 - Add orchestration and reproducible run order

STARTING STATE

Open the project workspace and your last saved evidence. Keep protected Review and any sealed incident/failure pack closed unless this phase explicitly unlocks it.

OUTCOME

Independently complete add orchestration and reproducible run order and produce reviewable evidence that proves the result, not merely the action.

TEACHER EXPLANATION

Reproducible run order puts sources/staging/tests before marts/publish. Failed tests must prevent downstream promotion.

TINY PATTERN - NOT YOUR FINAL SOLUTION

Represent task states and show that a partial run can resume from safe persisted models without claiming full publish.

BUILD THIS ARTIFACT

DAG/workflow + run log + dependency evidence

VALIDATE

Run dependency path end-to-end; prove failed task does not falsely advance downstream state; rerun only required work.

CHANGED / FAILURE TEST

Force an upstream task failure/partial run; prove downstream publish does not falsely succeed and rerun is bounded.

EVIDENCE + REVIEW ROUTE

Evidence ID: M18-P02-10-EVIDENCE. Save:

03_PROJECT_WORKSPACES/P02_ANALYTICS_ENGINEERING_PLATFORM/evidence/P02-10/attempt.json. P02 rubric -> exact phase criteria; review only after attempt evidence exists.

MASTER / MOVE-ON

L3 independent professional; no unresolved critical failure; reproducible artifact/evidence.

STOP - EXPLAIN BACK

Without reading this page, explain what changed in the system, what evidence proves it, one way it could be wrong, and what your changed/failure test taught you. Next: P02-11.

P02-11 - Inject duplicate/relationship/partial-run failures

STARTING STATE

Open the project workspace and your last saved evidence. Keep protected Review and any sealed incident/failure pack closed unless this phase explicitly unlocks it.

OUTCOME

Independently complete inject duplicate/relationship/partial-run failures and produce reviewable evidence that proves the result, not merely the action.

TEACHER EXPLANATION

Failure injection is where the model proves itself: duplicate dimension, orphan foreign key and partial run test different controls.

TINY PATTERN - NOT YOUR FINAL SOLUTION

Use each supplied failure separately; preserve first failed test/result, repair one cause, rerun all affected controls.

BUILD THIS ARTIFACT

first-failure evidence + root_cause.md + repair + fresh retry evidence

VALIDATE

Save first failing evidence before repair; identify root cause; apply smallest safe repair; fresh retry must pass all prior controls.

CHANGED / FAILURE TEST

MANDATORY SUPPLIED FAILURE: capture first failure, diagnose, repair, fresh retry; do not overwrite original evidence.

EVIDENCE + REVIEW ROUTE

Evidence ID: M18-P02-11-EVIDENCE. Save:

03_PROJECT_WORKSPACES/P02_ANALYTICS_ENGINEERING_PLATFORM/evidence/P02-11/attempt.json. P02 rubric -> exact phase criteria; review only after attempt evidence exists.

MASTER / MOVE-ON

L3 independent professional; no unresolved critical failure; reproducible artifact/evidence.

STOP - EXPLAIN BACK

Without reading this page, explain what changed in the system, what evidence proves it, one way it could be wrong, and what your changed/failure test taught you. Next: P02-12.

P02-12 - Document lineage, runbook, limitations and evidence

STARTING STATE

Open the project workspace and your last saved evidence. Keep protected Review and any sealed incident/failure pack closed unless this phase explicitly unlocks it.

OUTCOME

Independently complete document lineage, runbook, limitations and evidence and produce reviewable evidence that proves the result, not merely the action.

TEACHER EXPLANATION

Lineage and limitations make the platform reviewable. State what is simulated/unexecuted instead of decorating the README with claims.

TINY PATTERN - NOT YOUR FINAL SOLUTION

Evidence index links source -> model -> test -> reconciliation -> run log -> incident repair.

BUILD THIS ARTIFACT

README.md + runbook.md + evidence_index.md

VALIDATE

Reviewer can locate every evidence artifact and challenge one design trade-off; learner explains limitation and recovery path without reading a script.

CHANGED / FAILURE TEST

PHASE CHECK P02-12: one changed-input or broken-assumption test must be run and documented before mastery.

EVIDENCE + REVIEW ROUTE

Evidence ID: M18-P02-12-EVIDENCE. Save:

03_PROJECT_WORKSPACES/P02_ANALYTICS_ENGINEERING_PLATFORM/evidence/P02-12/attempt.json. P02 rubric -> exact phase criteria; review only after attempt evidence exists.

MASTER / MOVE-ON

L3 independent professional; no unresolved critical failure; reproducible artifact/evidence.

STOP - EXPLAIN BACK

Without reading this page, explain what changed in the system, what evidence proves it, one way it could be wrong, and what your changed/failure test taught you. Next: P02-13.

P02-13 - Clean-clone build and analytics-platform defense

STARTING STATE

Open the project workspace and your last saved evidence. Keep protected Review and any sealed incident/failure pack closed unless this phase explicitly unlocks it.

OUTCOME

Independently complete clean-clone build and analytics-platform defense and produce reviewable evidence that proves the result, not merely the action.

TEACHER EXPLANATION

A clean build should not depend on your shell history or forgotten local files. The defense must connect models to business meaning.

TINY PATTERN - NOT YOUR FINAL SOLUTION

Start from a clean copy, use declared config only, reproduce controls, then explain one model, one failure and one trade-off.

BUILD THIS ARTIFACT

clean-clone transcript/log + frozen evidence bundle

VALIDATE

Rebuild from documented clean state using only declared dependencies/config; reviewer can reproduce declared outputs.

CHANGED / FAILURE TEST

PHASE CHECK P02-13: one changed-input or broken-assumption test must be run and documented before mastery.

EVIDENCE + REVIEW ROUTE

Evidence ID: M18-P02-13-EVIDENCE. Save:

03_PROJECT_WORKSPACES/P02_ANALYTICS_ENGINEERING_PLATFORM/evidence/P02-13/attempt.json. P02 rubric -> exact phase criteria; review only after attempt evidence exists.

MASTER / MOVE-ON

L3 independent professional; no unresolved critical failure; reproducible artifact/evidence.

STOP - EXPLAIN BACK

Without reading this page, explain what changed in the system, what evidence proves it, one way it could be wrong, and what your changed/failure test taught you. Next: P03-01.